

PROJECT NAME	BOM10	Revision History	Rev No:	D01		
CLIENT	DIGITAL CONNECTION	Prepared By	Date:	04/08/2025		
LOCATION	Mushe	Reviewed By	YY			
DISCIPLINE	MECHANICAL LOW SIDE	Approved by	YY			04/08/2025
DOCUMENT NO	SUD-DX-BOM10-SPEC-ME-68	Submitted by	GS/SA			
Note: Please indicate all details which needs to be mentioned as comments/ values/ figures etc in comments as per format given. All new comments should be added above the previous comments as per instructions and format given. If otherwise, comments will not be considered. All vendor comments shall be replied to a new comment by Consultant in same cell, with the following format: {DDMM/YYYY} @Name: Comment.						
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Technical Specification for Low Side Items						
Sr.No.	Description	Other Specification details/ Graphics	Compliance to Specification	Vendor name/ Vendor/ Remarks Enter as {DDMM/YYYY} @Name: Comment	SUDLOWS Consultant global Remarks if any: Enter as {DDMM/YYYY} @Name: and then add Comment	
25	Insulation					
A	General All tests like pressure testing should be complete and recorded and all systems approved by the Consultants before insulation is applied to the equipment and piping. Insulation material shall be supplied by approved manufacturers and shall be of the type specifically intended for the services specified. All insulation material shall be non-combustible in accordance with BS 476 Part 4, ISO 1182, IS 3808.		Compliant			
B	Duct Insulation with Nitrile Rubber Thermal insulation of ducting shall be done with Elastomeric Foam finished with factory laminated GIC cloth (except for exposed ducting) structure of density 60 to 90 kg/m ³ as per ASTM-D-1667 of Class 'O' insulation material. • Apply one coat of adhesive on the outer surface of the sheet metal ducting and one coat of adhesive on one surface of the insulation material. • Allow it to dry for 1 minute and stick the insulation on the sheet metal ducting by pressing neatly. • Ensure that there is no air bubble entrapped or non-uniform fixing of the insulation.		Partially Compliant			
C	Acoustic insulation thickness shall be 25mm • The joints shall be sealed with 50x2mm of self-adhesive nitrile rubber tape. Chilled Water Pipe Insulation With Nitrile Rubber (Indoor) Wherever specified in BOQ, the thermal insulation of all chilled water pipes installed indoor shall be done with Class O closed cell elastomeric nitrile rubber insulation of density 60kg/m ³ . The nitrile rubber insulation shall be factory laminated GC Cloth of 0.18 mm thickness and coated with acrylic polymers or blends on one side and with self-adhesive backing on the other. The joints shall be sealed with 50x6mm of self-adhesive nitrile rubber tape.		Compliant			
D	Chilled Water Pipe Insulation With Polyurethane Foam (Outdoor & Shafts): Wherever specified BOQ, the thermal insulation of chilled water piping shall be done with fire retardant and self-extinguishing type closed cell Polyurethane Foam material of density not less than 40kg/m ³ as per the recommended thickness below. The step-by-step procedure for carrying out the insulation work shall be as follows: • Apply one coat of adhesive on the outer surface of the piping and one coat of adhesive on one surface of the insulation material. • Allow it to dry for 1 minute and stick the insulation on the piping by pressing neatly • Ensure that there is no air bubble entrapped or non-uniform fixing of the insulation. • The joints shall be sealed with 50mm of self-adhesive tape. The insulation over pipes which are installed outdoors and in shafts shall be finished with vapour barrier and cladding with ZIG A sheet.		Compliant			
E	Refrigerant Piping Insulation Material shall be Class O Nitrile Rubber insulation. Thickness of the insulation for liquid and gas pipe of DX system, VRF system will be as per manufacturer The properties shall be as follow: • Thermal conductivity of Elastomeric Nitrile rubber shall not exceed 0.035 W/m ² K at an average temperature of 0 deg C. • The insulation shall have fire performance such that it passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category. • Water vapor permeability shall not exceed 0.017 Perm inch (2.48 x 10 ⁻¹⁴ Kg/m.s.Pa), i.e. Moisture Diffusion Resistance Factor or 'μ' value should be minimum 7000. Application shall be as follows: • Clean the outer surface of refrigerant copper piping. Insert the pipes in tubular insulation. • Join two ends of tubular insulation using suitable adhesive. Tape the joints with masking tapes of the same material. • All outdoor piping to be protected with (For ref. Piping.) Woven Fiberglass cloth, 7 mil thickness and 200 gm weight, with factory laminated, self-adhesive backing should be used • This insulation has to be finished with 2 coats of UV painting.		Compliant			
F	Acoustic Insulation Flexible Open-cell Nitrile Rubber Elastomeric at every outlet of FCU, TFAHU up to 3m of length. a. Manufacturers: • Subject to compliance with requirements, the product should be selected only one from approved makes for entire project. (Refer BOQ). • Special Approval: Subject to compliance with requirements, the Owner's representative / Consultant have the authority to propose/approve the manufacturer (those not listed on approved make) on later stage of the project if warranted. b. Materials: Open-cellular polyethylene thermal plastic, preformed sheet insulation. • Thickness: 25-mm thick. • Thermal Conductivity (k-Value): 0.05 W/m ² °K at 24 °C mean temperature. • Density: 140 – 180 kg/cu. m • Fire-Hazard Classification: Conform to Class 1 rating for surface spread of Flame as per BS 476 Part 7. • Adhesive: As recommended by insulation manufacturer.		Compliant			
G	Under Deck Insulation The under-deck of the ceiling in data center hall/rooms as defined shall be provided with 25mm thick Nitrile rubber class 'O' multi-layer laminate and backed with a Aluminum foil on the lower surface and adhesive on upper surface. Under deck insulation shall be applied as follows: • Clean the surface and make it free from dust and loose particles by using a industrial type vacuum cleaner. • The insulation sheets shall be placed in position and compressed with uniform pressure to remove any trapped air bubbles. Stretching of insulation sheet shall be permitted to achieve a good bond. In case some air bubbles are noticed after pasting the sheet to the floor, the entire sheet shall be removed, and a new sheet pasted. • All longitudinal and transverse joints shall be sealed with 3mm thick and 50mm width self-adhesive class 'O' tape. • Insulation sheets shall be finished with SS wire mesh with screwing at 1mtr x 1 mtr interval		Compliant			
H	Under Floor Insulation The under-raised floor of the data center hall/Electrical rooms shall be provided with 16mm thick Nitrile rubber class 'O' with a Aluminum foil on the top surface and adhesive on lower surface. Under raised floor insulation shall be applied as follows: • Clean the surface and make it free from dust and loose particles by using a industrial type vacuum cleaner. • The insulation sheets shall be placed in position and compressed with uniform pressure to remove any trapped air bubbles. Stretching of insulation sheet shall be permitted to achieve a good bond. In case some air bubbles are noticed after pasting the sheet to the floor, the entire sheet shall be removed, and a new sheet pasted. • All longitudinal and transverse joints shall be sealed with 3mm thick and 50mm width self-adhesive class 'O' tape. • Insulation sheets shall be finished with wire mesh with screwing at 1mtr x 1 mtr interval • Insulation shall be applicable on the floor of data hall/Electrical rooms and the perimeter of the exposed walls upto the height of the raised floor for data hall/Electrical rooms, perimeter of the adjacent unconditioned spaces below raised floor upto the raised floor height for data hall/Electrical rooms.		Compliant			

